



THE GREEN HUB

TOGETHER FOR A GREENER TOMORROW



THE GREEN HUB

A Smart & Sustainable Community Center



Renewable Energy



Smart Technology



Community Impact



MBA Team Final Project — Group 5



RBC MBA — Project Management



Instructor: Dr. Ehab EL Nezamy



July 6, 2026

About

The Green Hub is a community center project that runs entirely on renewable energy and combines construction, engineering, and sustainability.



It represents the successful integration of professional project management principles with sustainable development practices — delivering measurable environmental, social, and economic value by combining renewable energy, smart technologies, and environmentally responsible design to inspire greener communities and foster a culture of green living.



“The Green Hub is a living embodiment of how professional project management converges with a vision of environmental sustainability to achieve a real, lasting impact on the community.”



Renewable Energy

Solar PV, rainwater harvesting, smart systems



Community Impact

Education, coworking, training programs



LEED Certified

Environmentally responsible design standard



Project Team

**THE GREEN HUB**

Sustainable Community Center

**Dr. Ehab EL Nezamy**

Instructor / Advisor

**Armia Kamal**

Project Manager

**Mohamed Ahab**

Planning & Scheduling Lead

**Michael Atef**

Scope & Risk Officer

**Salwa Michel**

Finance & Partners Officer

**Shehab Badr**

QA & Governance Officer

**Amira Gomaa**

QA & Audit Assistant



About The Green Hub

The Green Hub is a sustainable community center powered by renewable energy and smart technologies.

It combines green construction, environmental education, and community services to create long-term environmental, social, and economic value.

Key Focus Areas



**Renewable
Energy**



**Smart
Technology**



**Community
Impact**



**Sustainable
Design**





PROJECT SCOPE

Design and deliver The Green Hub, a sustainable community center powered by renewable energy and smart technologies.



INCLUDES

- Sustainable building design and construction
- Renewable energy systems
- Smart building management system (BMS)
- Community and educational facilities
- Testing, commissioning, and project handover



EXCLUDES

- Future building expansions
- Operations and facility management after handover

SCOPE DELIVERABLES



SUSTAINABLE FACILITY

Modern, energy-efficient community center



RENEWABLE ENERGY

Integrated solar PV and clean energy systems



SMART TECHNOLOGY

Smart building management and automation



COMMUNITY SERVICES

Spaces and programs for community development



EDUCATION & AWARENESS

Environmental education and training programs



ENVIRONMENTAL IMPACT

Reduced carbon footprint and resource conservation



5 PROJECT SCHEDULE & WBS

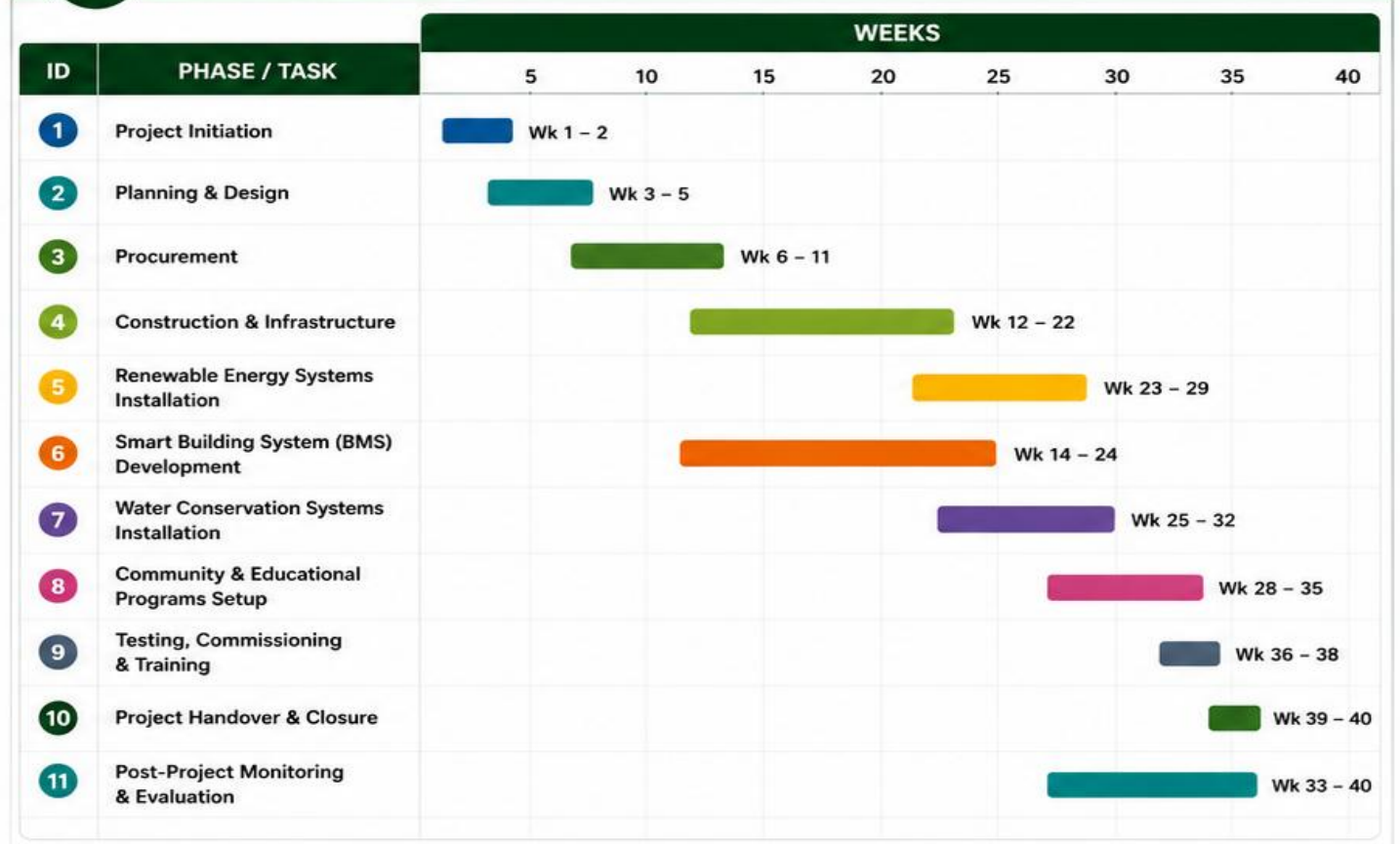
 Timeline: 40 Weeks | Start Date: 1 January 2026 | Finish Date: 2 October 2026



WORK BREAKDOWN STRUCTURE (WBS)



GANTT CHART (40 WEEKS)



KEY MILESTONES

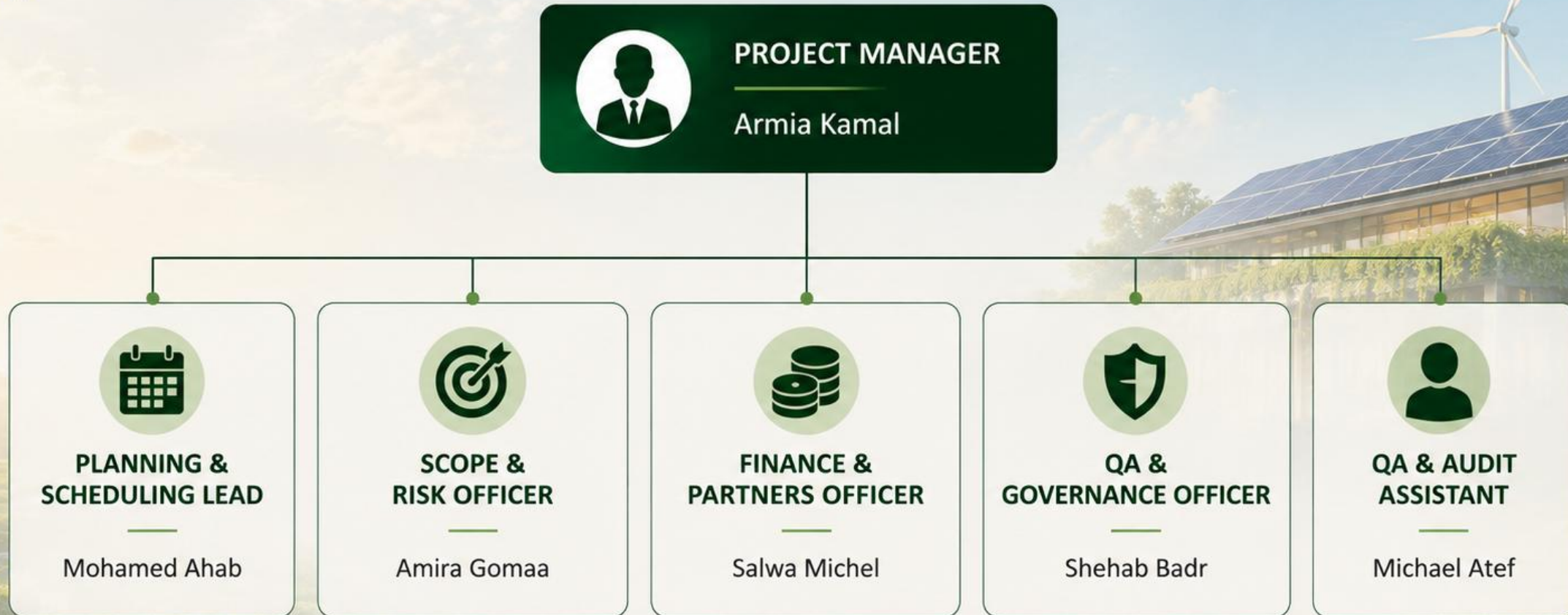
-  Design Approval Wk 5
-  Construction Completion Wk 22
-  Systems Commissioning Wk 38
-  Project Handover Wk 40



ORGANIZATIONAL BREAKDOWN STRUCTURE (OBS)



Per the instructor's rule: PM at the top, same horizontal level = same authority, titles not names.



7 BUDGET SUMMARY

Timeline: 40 Weeks | Start Date: 1 January 2026 | Finish Date: 2 October 2026



THE GREEN HUB
Sustainable Community Center

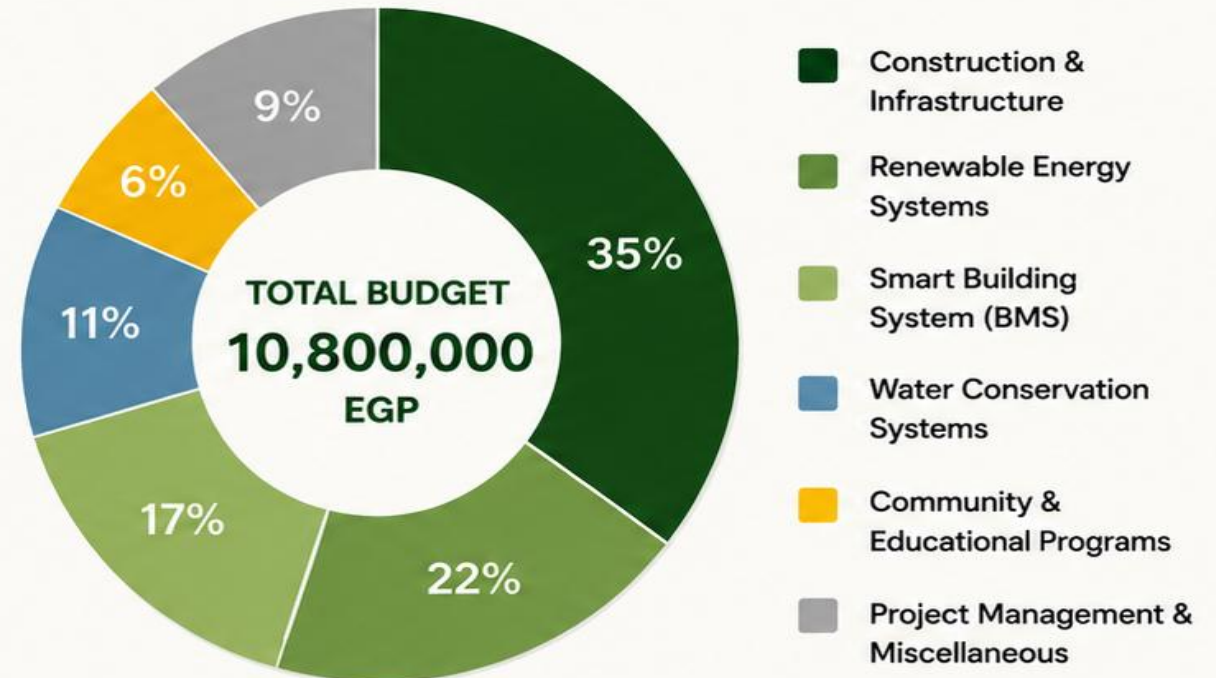


COST BREAKDOWN

CATEGORY		AMOUNT (EGP)	% OF TOTAL
	Construction & Infrastructure	3,800,000	35%
	Renewable Energy Systems	2,400,000	22%
	Smart Building System (BMS)	1,800,000	17%
	Water Conservation Systems	1,200,000	11%
	Community & Educational Programs	600,000	6%
	Project Management & Miscellaneous	1,000,000	9%
TOTAL PROJECT BUDGET		10,800,000	100%



BUDGET DISTRIBUTION





STAKEHOLDER REGISTER

STAKEHOLDER	ROLE / INTEREST	INFLUENCE (POWER)	INTEREST (LEVEL)	ENGAGEMENT STRATEGY
 Project Sponsor (Municipality)	Provides funding and strategic direction	High	High	 Manage Closely
 Project Manager & Team	Responsible for planning, execution and delivery	High	High	 Manage Closely
 Local Community & Users	Beneficiaries of the center and programs	Low	High	 Keep Informed
 Government & Regulatory Bodies	Approvals, permits and compliance	High	Medium	 Keep Satisfied
 Suppliers & Contractors	Provide materials, equipment and services	Medium	Medium	 Keep Satisfied
 Investors & Donors	Provide financial support and resources	High	High	 Manage Closely
 Educational & Training Partners	Support environmental education programs	Low	Medium	 Keep Informed
 Environmental Organizations	Support sustainability and green initiatives	Medium	High	 Keep Informed



STAKEHOLDER POWER-INTEREST MATRIX

High



INTEREST

Low



KEEP INFORMED

- Local Community & Users
- Educational & Training Partners
- Environmental Organizations



MANAGE CLOSELY

- Project Sponsor (Municipality)
- Project Manager & Team
- Investors & Donors



MONITOR

- (No major stakeholders in this quadrant)



KEEP SATISFIED

- Government & Regulatory Bodies
- Suppliers & Contractors

POWER (INFLUENCE)

Low

High



PROJECT SCHEDULE (GANTT CHART)

📅 Timeline: 40 Weeks | 📅 Start Date: 1 January 2026 | 📅 Finish Date: 4 October 2026



THE GREEN HUB

Sustainable Community Center

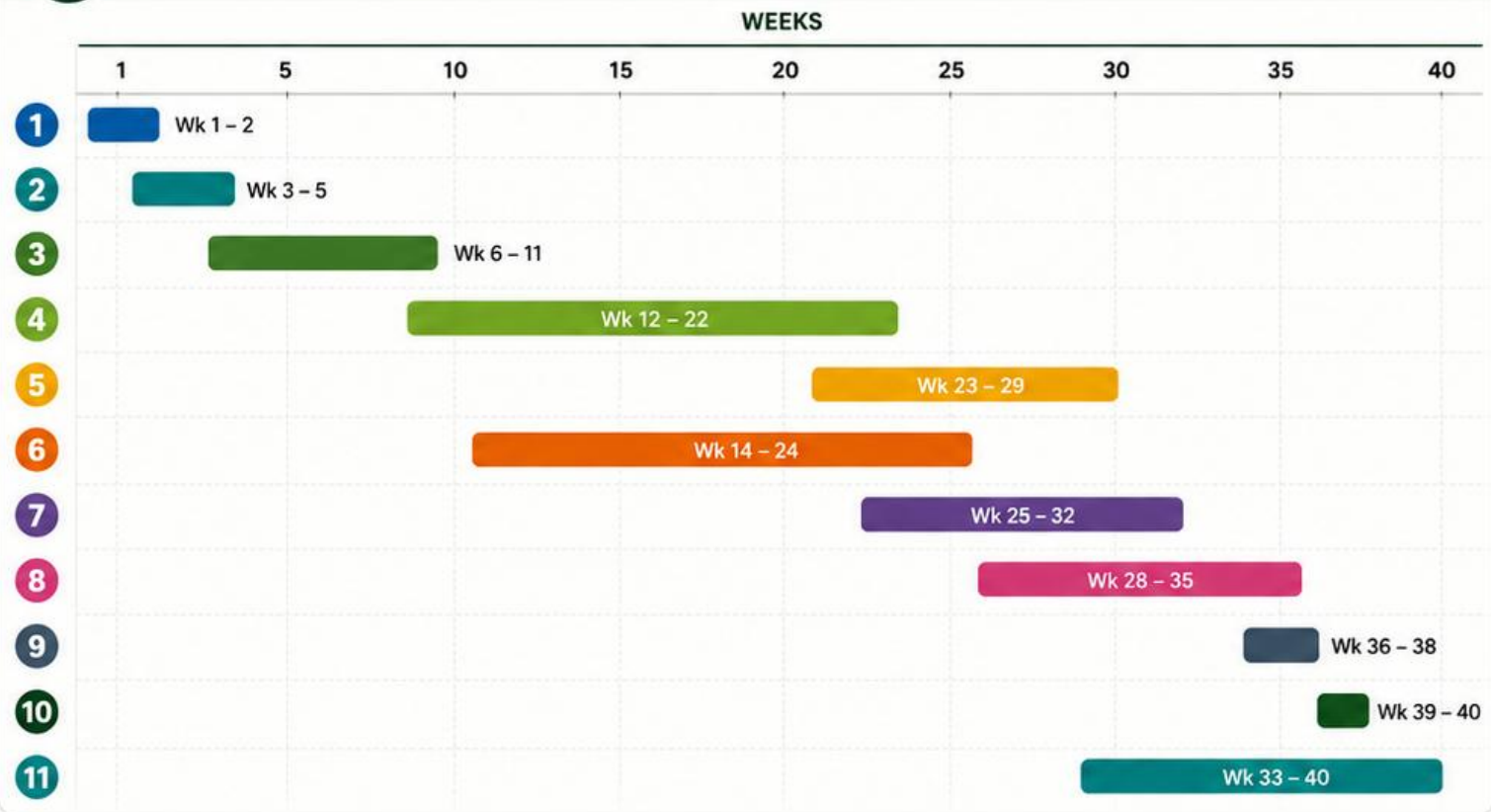


PROJECT PHASES & TASKS

ID	PHASE / TASK	DURATION (WEEKS)	START	FINISH
1	 Project Initiation	2	Wk 1	Wk 2
2	 Planning & Design	3	Wk 3	Wk 5
3	 Procurement	6	Wk 6	Wk 11
4	 Construction & Infrastructure	11	Wk 12	Wk 22
5	 Renewable Energy Systems Installation	7	Wk 23	Wk 29
6	 Smart Building System (BMS) Development	11	Wk 14	Wk 24
7	 Water Conservation Systems Installation	8	Wk 25	Wk 32
8	 Community & Educational Programs Setup	8	Wk 28	Wk 35
9	 Testing, Commissioning & Training	3	Wk 36	Wk 38
10	 Project Handover & Closure	2	Wk 39	Wk 40
11	 Post-Project Monitoring & Evaluation	8	Wk 33	Wk 40
TOTAL DURATION		40 WEEKS		



PROJECT TIMELINE (40 WEEKS)



KEY MILESTONES



Design Approval
Wk 5



Construction
Completion
Wk 22



Systems
Commissioning
Wk 38



Project Handover
Wk 40



THE GREEN HUB

TOP 8 PROJECT RISKS & MITIGATION PLAN



Identify. Assess. Mitigate.
Managing Risks. Delivering Success.

#	RISK	PROBABILITY	IMPACT		OWNER	MITIGATION STRATEGY	STRATEGY TYPE
			💰 COST (EGP)	🕒 TIME (WEEKS)			
 1	Building Permit Delay Delay in obtaining construction approval.	High	900,000	4 wks	 Design Lead	Submit permit at 70% design and appoint a permitting consultant.	 MITIGATE
 2	Late Delivery of Solar Equipment Delay in PV panels & inverters (shipping / customs).	High	1,400,000	5 wks	 Procurement Manager	Order early in Week 14; dual-source suppliers; include late-delivery penalty clause.	 TRANSFER
 3	Currency Fluctuation EGP devaluation increases cost of imported equipment.	High	2,700,000	0 wks	 Finance Manager	Fix prices in EGP; pay 50% advance; cap escalation at 5%.	 MITIGATE
 4	Contractor Underperformance Main contractor may underperform or default.	Medium	3,600,000	8 wks	 Project Manager	Prequalify bidders; 10% performance bond; 5% retention; milestone payments.	 MITIGATE
 5	Scope Creep New requests change scope, impacting time & cost.	High	1,300,000	3 wks	 Project Manager	Formal change control – no change without Steering Committee approval.	 MITIGATE
 6	Utility Connection Delay Delay in grid connection / net-metering approval.	Medium	450,000	4 wks	 Sustainability Engineer	Apply in Week 14; use temporary generator for testing can continue.	 MITIGATE
 7	Material Price Increase Escalation in cement and steel prices.	High	1,800,000	0 wks	 Procurement Manager	Bulk-buy at award; fixed-price supply contracts; 7% contingency.	 SHARE
 8	Safety Incidents Accidents on site (e.g. fall from roof / PV works).	Medium	1,000,000	2 wks	 QHSE Officer	Insurance; permit-to-work; edge protection; weekly toolbox talks.	 MITIGATE



TOTAL POTENTIAL COST IMPACT
17,850,000 EGP



TOTAL POTENTIAL DELAY
43 WEEKS



TOTAL MITIGATION COST
3,300,000 EGP

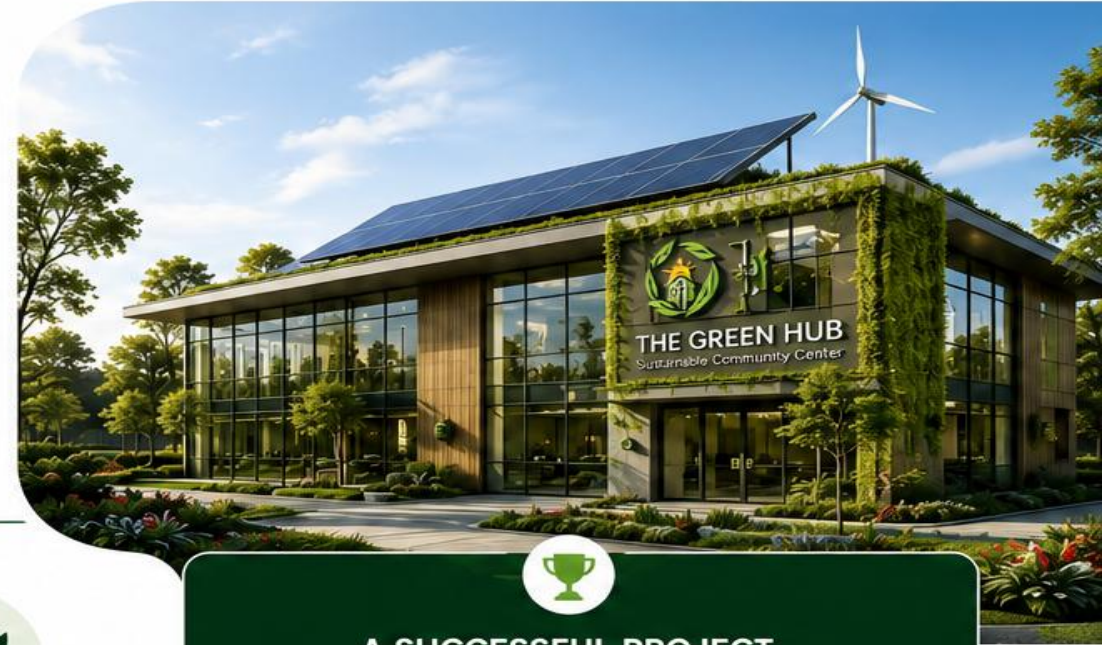


OUR GOAL
**ZERO SURPRISES.
ON TIME. ON BUDGET.**



“

The **Green Hub** is a living example of how professional project management and sustainable innovation come together to create **lasting value** for people and the planet.



KEY TAKEAWAYS



ON TIME

Delivered within the **40-week timeline** with a structured plan and effective execution.



ON BUDGET

Completed within the approved budget of **10,800,000 EGP** with strong cost control.



HIGH QUALITY

Achieved the highest quality standards in construction, systems, and sustainability.



COMMUNITY IMPACT

Delivering long-term environmental, social, and economic value to the community.



SUSTAINABLE FUTURE

Built to operate efficiently, reduce environmental impact, and inspire a greener tomorrow.



A SUCCESSFUL PROJECT. A SUSTAINABLE LEGACY.

Thank you to our team, stakeholders, partners, and supporters for their dedication and collaboration in making The Green Hub a reality.

*Together, we build
a better future.*



PROJECT IMPACT

Beyond Construction: Long-Term Sustainable Value

A practical impact on the environment, the community, and future operations.



**Sustainable
Impact**



Reduce Carbon Emissions

Lower footprint through efficient design and clean energy.



Renewable Energy

Solar PV and smart controls support green operations.



Water Conservation

Rainwater harvesting and recycling reduce waste.



Community Development

Education, training, and shared spaces create social value.



Long-term Sustainability

A scalable model with lasting social and economic benefit.



Thank You

Together for a greener tomorrow

The Green Hub — MBA Team Final Project, Group 5

Salwa Michel - Armia Kamal - Mohamed Ahab - Michael Atef - Shehab Badr - Amira Gomaa

 **THE GREEN HUB**
SUSTAINABLE COMMUNITY CENTER